Tanmay Pawar

+91-7507531801 tanmaymsp@gmail.com LinkedIn GitHub

Professional Summary

Machine Learning Engineer with M.Tech in AI from IIT Patna, experienced in developing AI applications using LLMs, RAG, computer vision, and deep learning. Built OCR-driven evaluation workflows, fine-tuned LLMs, and developed ML solutions with a focus on reliable and measurable performance.

Education

Indian Institute of Technology Patna — *Patna, Bihar*
M.Tech in AI | CGPA: 8.94 — *Aug. 2024 – June 2026*

AISSMS College of Engineering, Pune — *Pune, Maharashtra*
B.Tech in ECE | CGPA: 9.37 — *Sept. 2018 – April 2022*

Technical Skills

- **Languages:** Python, SQL, LookML
- **ML & AI:** PyTorch, Hugging Face, Transformers, LangChain, Google GenAI SDK
- **Data & Databases:** Pandas, NumPy, Matplotlib, Seaborn, Pydantic, BigQuery, FAISS
- **Workflow, Cloud & DevOps:** Airflow, GCP, Docker, GitHub Actions, Git
- **Core Competencies:** Machine Learning, Deep Learning, Generative AI, NLP, LLMs, Computer Vision

Experience

Warner Bros. Discovery — *Oct. 2025 – July 2026*
Machine Learning Engineer Intern — *Bangalore, India*

- Developed a burnt-in subtitle removal component using OCR-detected text to filter generated subtitles, reducing falsely inflated Word Error Rate by ≈ **23%** on heavily subtitled videos.
- Built a **Gemini**-assisted classification layer on OCR outputs to classify subtitles, credits and other on-screen text, enabling reliable filtering within caption evaluation pipelines.
- Implemented multi-class classification matrices for caption positioning in video under highly imbalanced data.
- Implemented **GitHub Actions** workflow to build and publish a versioned Python package to GCP Artifact Registry.

Volkswagen Group Technology Solutions India — *Aug. 2022 – Jun. 2024*
Software Engineer — *Pune, India*

- Delivered a predictive-maintenance PoC for equipment health monitoring: benchmarked XGBoost, LSTM and GRU for remaining-useful-life regression on EMA-smoothed sensor data, and built Autoencoder degradation-onset detection reaching ≈ 0.92 f1-score at the 95th-percentile healthy reconstruction threshold.
- Implemented it's application with **FastAPI** backend connected to streamlit UI for pilot project use.

Intangles Labs Pvt. Ltd. — *Oct. 2021 – Apr. 2022*
ML Project Intern — *Pune, India*

- Worked on Driver-Monitoring System for, on-edge face detection using YOLOv4-tiny fine-tuned for a custom class.
- Extended to embedding-based driver identification using FaceNet with similarity search, achieving ≈ **93%** accuracy on balanced class distribution.

Projects

RAG-Augmented Golden Response Generation for Empathetic Dialogue — *IIT Patna*

- Built domain-specific RAG pipeline grounded in psychology literature with COMET-ATOMIC for query expansion, FAISS-based retrieval, and cross-encoder reranker.
- Generated high-quality empathetic responses with LLaMA-3.2-3B-Instruct, outperforming crowd-sourced ground truth (**4.53** vs 3.08) per a custom LLM-as-judge framework; used outputs to build an dataset.
- Performed LoRA-based supervised fine-tuning with A/B testing on generated vs. crowd-sourced data, surpassing the crowd-data baseline by +**1.16 points** on empathic quality.

– GPT-2 Playground — [Live Demo]

- Re implemented GPT-2 (124M) in PyTorch using pretrained weights.
- Built interactive demo for token-level generation under multiple **decoding strategies** (Top-K, Top-P, Min-P, temperature) and **constrained decoding** modes (digits-only, alphabetic, custom vocabulary, forbidden words). **Speculative Decoding** demo for faster inference.
- Deployed an interactive text generation demo on Hugging Face Spaces using Dockerized inference, enabling exploration of model generation dynamics.

AQI-Pulse: MLOps Pipeline for AQI forecasting — [GitHub]

- Built and deployed an end-to-end MLOps system forecasting air-quality time series on GCP Compute Engine with a champion–challenger architecture, orchestrated via Airflow, tracked with MLflow, and containerized with Docker.
- Automated accuracy monitoring, triggering model retraining and challenger promotion on held-out validation set

Achievements & Certifications

- **AWS AI/ML Scholar 2022–23** — Selected among 500 students worldwide for the AWS × Udacity scholarship. *2022*
- **GATE 2024 Qualified** in Data Science and Artificial Intelligence (DA) AIR 724, 98.15 percentile.